Due by2359 hours on 20260817

Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **2C** **2026** Fall/Spring/Summer I/**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	4	1	1							6	
1b	3	3								6	
1c	1	4								5	
2a	6									6	
2b	6									6	
2c	2	3								5	
3a	1	1	4							6	
3b	1	1								2	
3c	9									9	
								Total		51	

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \|, \perp, \setminus, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by striaght lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

Instructions:

R Do not use decimals anywhere except if explicitly noted in the question.

R Do not make any extraneous marks anywhere on the exam.

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points for otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

Instructions:

Your J-number = 00499317

Keep every 0 in your J-number and replace every non-zero number in your J-number by 1 to get

Your (truth-value) **T-number** := 00111111

Define $\forall k \in 1..8$,

T_k ≔ the k[th] entry in your **truth-value J-number** counting from the left. Therefore,

$T_1 \coloneqq 0$

$T_2 \coloneqq 0$

$T_3 \coloneqq 1$

$T_4 \coloneqq 1$

$T_5 \coloneqq 1$

$T_6 \coloneqq 1$

$T_7 \coloneqq 1$

$T_8 \coloneqq 1$

Next construct an expression $G(w, x, y, z)$ as follows:

$$\alpha(T_1, T_2, T_3, T_4)$$

$$= \quad (T_1)(w) + (T_2)(x') + (T_3)(y) + (T_4)(z')$$

$$= \quad (0)(w) + (0)(x') + (1)(y) + (1)(z')$$

$$= \quad y + z'$$

$$\beta(T_3, T_4, T_5, T_6)$$

$$= \left(\left((T_3)w\right)'\right)\left(\left((T_4)x\right)'\right)\left(\left((T_5)y\right)'\right)\left(\left((T_6)z\right)'\right)'$$

$$= ((1)w)'((1)x)'((1)y)'((1)z)''$$

$$= (w')(x')(y')(z')'$$

$$= w + x + y + z$$

$$\gamma(T_5, T_6, T_7, T_8)$$

$$= (T_5)(w') + (T_6)(x) + (T_7)(y') + (T_8)(z)$$

$$= (1)(w') + (1)(x) + (1)(y') + (1)(z)$$

$$= w' + x + y' + z$$

=

$$G(w, x, y, z)$$

$$= \quad \Big(\alpha(T_1, T_2, T_3, T_4)\Big)\Big(\beta(T_3, T_4, T_5, T_6)\Big)\Big(\gamma(T_5, T_6, T_7, T_8)\Big)$$

$$= \quad \Big(y + z'\Big)\Big(w + x + y + z\Big)\Big(w' + x + y' + z\Big)$$

=

6 (1a) Find $\varphi(p,q,r)$ in terms of: p,q and r if $TT\left(\varphi(p,q,r)\right)$ is given below. Find corresponding expressions $F(A,B,C)$ in $\underline{\underline{Sets}}$ and $E(x,y,z)$ in $\underline{\underline{BA}}$ on the facing page and record the answers on this page.

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	$T_1 = 0$	0	0	0	0	0	0	0	0
0	0	1	$T_2 = 0$	0	0	0	0	0	0	0	0
0	1	0	$T_3 = 1$	0	0	0	0	0	1	0	0
0	1	1	$T_4 = 1$	0	0	0	0	1	0	0	0
1	0	0	$T_5 = 1$	0	0	0	1	0	0	0	0
1	0	1	$T_6 = 1$	0	0	1	0	0	0	0	0
1	1	0	$T_7 = 1$	0	1	0	0	0	0	0	0
1	1	1	$T_8 = 1$	1	0	0	0	0	0	0	0

4(i) $\varphi(p,q,r)$

$= (p \wedge q \wedge r) \vee (p \wedge q \wedge (\neg r)) \vee (p \wedge (\neg q) \wedge r)$
$\vee (p \wedge (\neg q) \wedge (\neg r)) \vee ((\neg p) \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r))$

1 (ii) $F(A,B,C)$

$= (A \cap B \cap C) \cup (A \cap B \cap (C^c)) \cup (A \cap (B^c) \cap C)$
$\cup (A \cap (B^c) \cap (C^c)) \cup ((A^c) \cap B \cap C) \cup ((A^c) \cap B \cap (C^c))$

1 (iii) $E(x,y,z)$

$= xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'$

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1a

$\varphi(p,q,r)$

$= \varphi_1 \vee \varphi_2 \vee \varphi_3 \vee \varphi_4 \vee \varphi_5 \vee \varphi_6$

$= ((\neg p) \wedge q \wedge (\neg r))$
$\vee ((\neg p) \wedge q \wedge r)$
$\vee (p \wedge (\neg q) \wedge (\neg r))$
$\vee (p \wedge (\neg q) \wedge r)$
$\vee (p \wedge q \wedge (\neg r))$
$\vee (p \wedge q \wedge r)$

$F(A,B,C)$

$= ((A^c) \cap B \cap (C^c))$
$\cup ((A^c) \cap B \cap C)$
$\cup (A \cap (B^c) \cap (C^c))$
$\cup (A \cap (B^c) \cap C)$
$\cup (A \cap B \cap (C^c))$
$\cup (A \cap B \cap C)$

$E(x,y,z)$

$= x'yz'$
$+ x'yz$
$+ xy'z'$
$+ xy'z$
$+ xyz'$
$+ xyz$

6 (1b) Use the consensus method to find the following for your answer from (1a):

$$E(x, y, z) \quad = \quad xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'$$

3 (i) the prime implicants of E: $PI\ (E) = \{x, y\}$

3 (i) the minimal sum-of-products form: $MSOP\ (E) = x + y$

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1b.i.0

E

$$xyz + xyz' + xy'z + xy'z' + x'yz + x'yz'$$

$$= xy'z + xy'z' + x'yz + x'yz' + xy \qquad \text{Cns}(xyz,xyz')=xy$$

$$= x'yz + x'yz' + xy + xy' \qquad \text{Cns}(xy'z,xy'z')=xy'$$

$$= xy + xy' + x'y \qquad \text{Cns}(x'yz,x'yz')=x'y$$

$$= xy' + y \qquad \text{Cns}(xy,x'y)=y$$

$$= y + x \qquad \text{Cns}(xy',y)=x$$

1b.i.1

$(x \not\subseteq y)\ (\ y \not\subseteq x)\ \ \not\text{Cns}(x,y)$

$$PI(E) = \{x, y\}$$

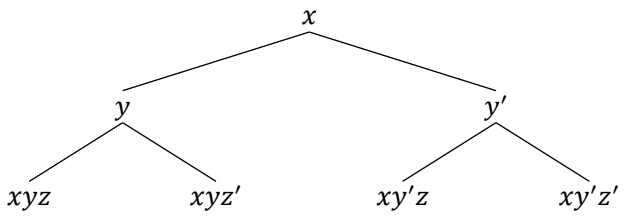
1b.ii.0

E

$$x = x(1)(1) = x(y+y')(z+z') = xyz(11)+xyz'(21)+xy'z+xy'z' = xy(11)+xy' = x = MSOP(E)$$
$$\quad +y \quad +(1)y(1) \quad +(x+x')y(z+z') \quad +xyz(12)+xyz'(22)+x'yz+x'yz' \quad +xy(12)+x'y \quad +y$$

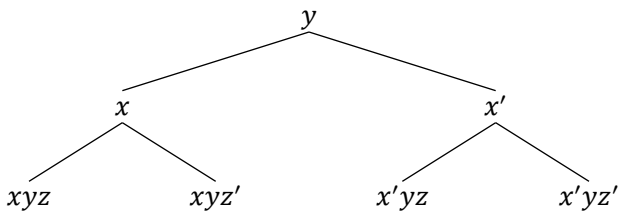
1b.ii.1

Tree



1b.ii.2

Tree



5 (1c) From (1b) $MSOP\ (E) = \ x + y$

1 (i) Find the version $\psi(p, q, r)$ of $MSOP\ (E)$ in PC

$$\psi(p, q, r) = \ p \vee q$$

4 (ii) **Prove in PC** that your answer from (1a) $\varphi(p, q, r) = \psi(p, q, r)$

$\varphi(p,q,r)$

$(p \wedge q \wedge r) \vee (p \wedge q \wedge (\neg r)) \vee (p \wedge (\neg q) \wedge r)$
$(p \wedge (\neg q) \wedge (\neg r)) \vee ((\neg p) \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r))$
$= \ (p \wedge (\neg q) \wedge r) \vee (p \wedge (\neg q) \wedge (\neg r)) \vee ((\neg p) \wedge q \wedge r)$
$((\neg p) \wedge q \wedge (\neg r)) \vee (p \wedge q)$
$= \ ((\neg p) \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r)) \vee (p \wedge q)$
$(p \wedge \neg q)$
$= \ (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$
$= \ (p \wedge \neg q) \vee q$
$= \ q \vee p$
$= \ p \vee q = \ \psi(p,q,r)$

1c.ii.0

$$\varphi(p,q,r)$$

$$((\neg p)\wedge q\wedge(\neg r))\vee((\neg p)\wedge q\wedge r)\vee(p\wedge(\neg q)\wedge(\neg r))$$
$$\vee(p\wedge(\neg q)\wedge r)\vee(p\wedge q\wedge(\neg r))\vee(p\wedge q\wedge r)$$

$$(p\wedge q\wedge(r\vee\neg r))\vee(p\wedge(\neg q)\wedge(r\vee\neg r))\vee((\neg p)\wedge q\wedge(r\vee\neg r))$$

$$(p\wedge q)\vee(p\wedge\neg q)\vee(\neg p\wedge q)$$

$$(p\wedge(q\vee\neg q))\vee(\neg p\wedge q)$$

$$p\vee(\neg p\wedge q)$$

$$(p\vee\neg p)\wedge(p\vee q)$$

$$p\vee q$$

$$\psi(p,q,r)$$

$$\varphi(p,q,r)=\psi(p,q,r)$$

6 (2a) For the expression from page 605 below:

$$G(w, x, y, z) = (y + z')(w + x + y + z)(w' + x + y' + z)$$

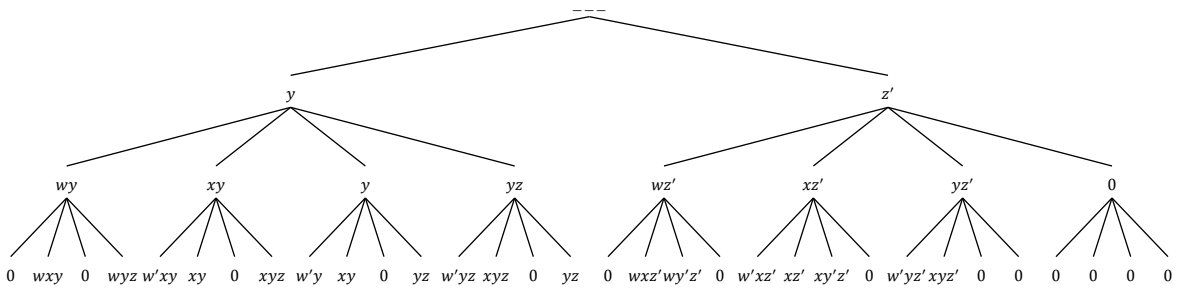
find a sum-of-products form, $SOP(G)$

$$SOP(G) = wy'z' + w'y + xy + xz' + yz$$

Show the tree on the facing side.

2a.0

Tree



2a.1

$G(w,x,y,z)$

$= 0+wxy+0+wyz+w'xy+xy+0+xyz+w'y+xy+0+yz+w'yz+xyz+0+yz+0+wxz'+wy'z'+0+w'xz'+xz'+xy'z'+0+w'yz'+xyz'+0+0+0+0+0+0$

$= \underline{wxy} + \underline{wyz} + \underline{w'xy} + xy + \underline{xyz} + w'y + yz + \underline{w'yz} + \underline{wxz'} + wy'z' + \underline{w'xz'} + xz' + \underline{xy'z'} + \underline{w'yz'} + \underline{xyz'}$

$= xy + w'y + yz + wy'z' + xz'$

$(xy \subseteq wxy,\ xy \subseteq w'xy,\ xy \subseteq xyz,\ xy \subseteq xyz'$

$(w'y \subseteq w'xy,\ w'y \subseteq w'yz,\ w'y \subseteq w'yz',\ yz \subseteq wyz$

$(yz \subseteq xyz,\ yz \subseteq w'yz,\ xz' \subseteq wxz',\ xz' \subseteq w'xz'$

$(xz' \subseteq xy'z',\ xz' \subseteq xyz')$

2a.2

$xy \supseteq w'y \quad w'y \supseteq xy$

$xy \supseteq yz \quad yz \supseteq xy$

$xy \supseteq wy'z' \quad wy'z' \supseteq xy$

$xy \supseteq xz' \quad xz' \supseteq xy$

$w'y \supseteq yz \quad yz \supseteq w'y$

$w'y \supseteq wy'z' \quad wy'z' \supseteq w'y$

$w'y \supseteq xz' \quad xz' \supseteq w'y$

$yz \supseteq wy'z' \quad wy'z' \supseteq yz$

$yz \supseteq xz' \quad xz' \supseteq yz$

$wy'z' \supseteq xz' \quad xz' \supseteq wy'z'$

2a.3

$$G(w,x,y,z)$$
$$= xy + w'y + yz + wy'z' + xz'$$
$$= SOP(G)$$

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6 (2b) For the expression below (**which is the same as in (2a)**):

$$G(w, x, y, z) = (y + z')(w + x + y + z)(w' + x + y' + z)$$

find the complete sum-of-products form, $CSOP(G)$

$$\begin{aligned} CSOP(G) \quad = \quad & w'x'yz' \\ & + \; w'x'yz \\ & + \; w'xy'z' \\ & + \; w'xyz' \\ & + \; w'xyz \\ & + \; wx'y'z' \\ & + \; wx'yz \\ & + \; wxy'z' \\ & + \; wxyz' \\ & + \; wxyz \end{aligned}$$

2b.0

$SOP(G)$

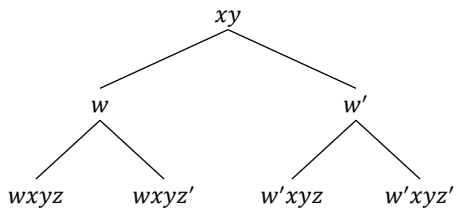
=	xy	$= (1)xy(1)$	$= (w+w')xy(z+z')$	$= wxyz+wxyz'+w'xyz+w'xyz'$
+	$w'y$	$= w'(1)y(1)$	$= w'(x+x')y(z+z')$	$= w'xyz+w'xyz'+w'x'yz+w'x'yz'$
+	yz	$= (1)(1)yz$	$= (w+w')(x+x')yz$	$= wxyz+wx'yz+w'xyz+w'x'yz$
+	$wy'z'$	$= w(1)y'z'$	$= w(x+x')y'z'$	$= wxy'z'+wx'y'z'$
+	xz'	$= (1)x(1)z'$	$= (w+w')x(y+y')z'$	$= wxyz'+wxy'z'+w'xyz'+w'xy'z'$

$CSOP(G)$

$$= w'x'yz + w'x'yz' + w'xy'z' + w'xyz + w'xyz'$$
$$+ \quad wx'y'z' + wx'yz + wxy'z' + wxyz + wxyz'$$

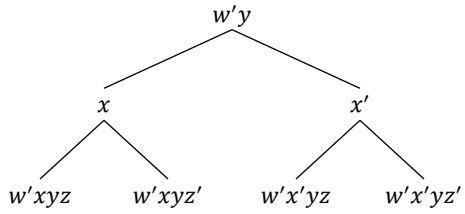
2b.1

Tree



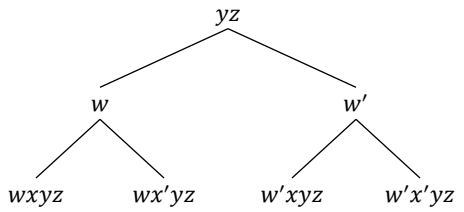
2b.2

Tree



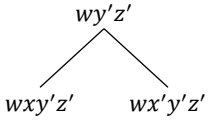
2b.3

Tree



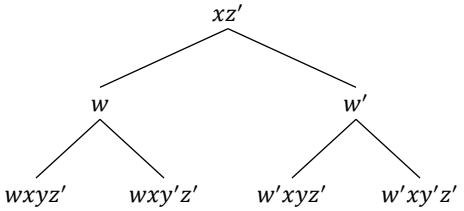
2b.4

Tree



2b.5

Tree



5 (2c) For the expression below **which is the same as in (2a) and (2b)**:

$$G(w, x, y, z) = (y + z')(w + x + y + z)(w' + x + y' + z)$$

use your answer from (2a) and (2b) to find the following:

$PI(G)$ and $MSOP\ (G)$

Answer from (2a) $SOP(G) = wy'z' + w'y + xy + xz' + yz$

Answer from (2b) $CSOP\ (G) = w'x'yz' + w'x'yz + w'xy'z' + w'xyz' + w'xyz + wx'y'z' + wx'yz + wxy'z' + wxyz' + wxyz$

2 (i) $PI(G) = \{w'y, wy'z', xy, xz', yz\}$

3 (ii) $MSOP\ (G) = w'y + wy'z' + xz' + yz$

2c.i.0

$$SOP(G) = wy'z' + w'y + xy + xz' + yz$$

$$PI(G) = \left\{w'y,\ wy'z',\ xy,\ xz',\ yz\right\}$$

2c.i.1

$$CSOP(G)$$

Cns($w'xy'z'$,$wxy'z'$)=$xy'z'$
Cns($w'x'yz'$,$w'x'yz$)=$w'x'y$
Cns($wx'y'z'$,$wxy'z'$)=$wy'z'$
Cns($w'x'yz$,$w'xyz$)=$w'yz$
Cns($w'xyz'$,$wxyz'$)=xyz'
Cns($wxyz'$,$wxyz$)=wxy
Cns($wx'yz$,$wxyz$)=wyz

Cns($w'xy'z'$,$w'xyz'$)=$w'xz'$
Cns($w'x'yz'$,$w'xyz'$)=$w'yz'$
Cns($wxy'z'$,$wxyz'$)=wxz'
Cns($w'x'yz$,$wx'yz$)=$x'yz$
Cns($w'xyz'$,$w'xyz$)=$w'xy$
Cns($w'xyz$,$wxyz$)=xyz

Cns($xy'z'$,xyz')=xz'
Cns($w'xz'$,wxz')=xz'
Cns($w'xy$,wxy)=xy
Cns($w'yz$,wyz)=yz

Cns($w'yz'$,$w'yz$)=$w'y$
Cns($w'x'y$,$w'xy$)=$w'y$
Cns(xyz',xyz)=xy
Cns($x'yz$,xyz)=yz

$$PI(G) = \left\{w'y,\ wy'z',\ xy,\ xz',\ yz\right\}$$

2c.i.2

$w'y \not\subseteq wy'z'$	$wy'z' \not\subseteq w'y$	~~Cns($w'y$,$wy'z'$)~~
$w'y \not\subseteq xy$	$xy \not\subseteq w'y$	~~Cns($w'y$,xy)~~
$w'y \not\subseteq xz'$	$xz' \not\subseteq w'y$	~~Cns($w'y$,xz')~~
$w'y \not\subseteq yz$	$yz \not\subseteq w'y$	~~Cns($w'y$,yz)~~
$wy'z' \not\subseteq xy$	$xy \not\subseteq wy'z'$	Cns($wy'z'$,xy) $\subseteq xz'$
$wy'z' \not\subseteq xz'$	$xz' \not\subseteq wy'z'$	~~Cns($wy'z'$,xz')~~
$wy'z' \not\subseteq yz$	$yz \not\subseteq wy'z'$	~~Cns($wy'z'$,yz)~~
$xy \not\subseteq xz'$	$xz' \not\subseteq xy$	~~Cns(xy,xz')~~
$xy \not\subseteq yz$	$yz \not\subseteq xy$	~~Cns(xy,yz)~~
$xz' \not\subseteq yz$	$yz \not\subseteq xz'$	Cns(xz',yz) $\subseteq xy$

2c.i.3

$$PI(G) = \{w'y,\ wy'z',\ xy,\ xz',\ yz\}$$

2c.ii.0

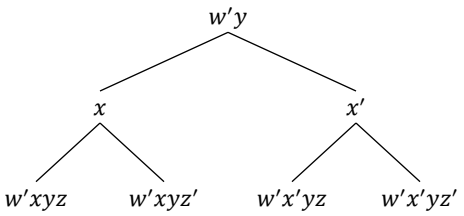
G

$= w'y$	$= w'(1)y(1)$	$= w'(x+x')y(z+z')$	$= w'xyz_{(1)}+w'xyz'_{(2)}+w'x'yz_{(3)}+w'x'yz'$
$+ wy'z'$	$= w(1)y'z'$	$= w(x+x')y'z'$	$= wxy'z'_{(4)}+wx'y'z'$
$+ xz'$	$= (1)x(1)z'$	$= (w+w')x(y+y')z'$	$= wxyz'+wxy'z'_{(4)}+w'xyz'_{(2)}+w'xy'z'$
$+ yz$	$= (1)(1)yz$	$= (w+w')(x+x')yz$	$= wxyz+wx'yz+w'xyz_{(1)}+w'x'yz_{(3)}$

$= w'y = MSOP(G)$
$= wy'z'$
$= xz'$
$= yz$

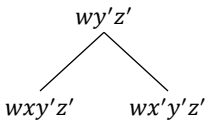
2c.ii.1

Tree



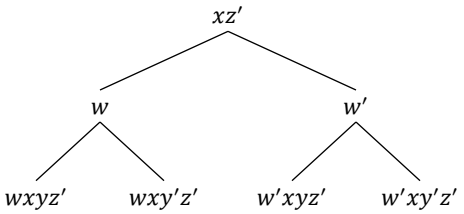
2c.ii.2

Tree



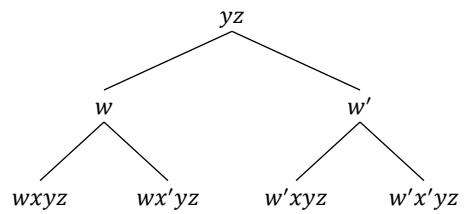
2c.ii.3

Tree



2c.ii.4

Tree



6 (3a) We define:

$$A := \ SOP\Big(G(w,x,y,z)\Big) = \ wy'z' + w'y + xy + xz' + yz$$

$$B \ = \ MSOP\Big(G(w,x,y,z)\Big) = \ w'y + wy'z' + xz' + yz$$

Copy your answers from (1a)

Compute:

1 (i) $A_S \ = \ 5$ $A_L \ = \ 11$ $B_S \ = \ 4$ $B_L \ = \ 9$

1 (ii) On the basis of your answers to (i), circle the correct choice below.

$Smplr\ (A,B)$ $Smplr(B,A)$ Neither

Why? (BL=9<11=AL) and (BS=4≤5=AS), so $Smplr(B,A)$ holds by definition.

4 (iii) Write down the version of $G(w,x,y,z)$ in PC below and find its **formation-tree on the facing side**.

Version of $G(w,x,y,z)$in PC

$= \ (r \vee \neg s) \wedge (p \vee q \vee r \vee s) \wedge (\neg p \vee q \vee \neg r \vee s)$

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3a.i.0

A_S

$= \mathrm{v}(\mathrm{Smnd}(A))$

$= \mathrm{v}(\{wy'z', w'y, xy, xz', yz\})$

$= 5$

3a.i.1

A_L

$= \mathrm{v}(\mathrm{MSLtr}(A))$

$= \mathrm{v}(\{w{:}1, w'{:}1, x{:}2, y{:}3, y'{:}1, z{:}1, z'{:}2\})$

$= 11$

3a.i.2

B_S

$= \mathrm{v}(\mathrm{Smnd}(B))$

$= \mathrm{v}(\{w'y, wy'z', xz', yz\})$

$= 4$

3a.i.3

B_L

$= \mathrm{v}(\mathrm{MSLtr}(B))$

$= \mathrm{v}(\{w{:}1, w'{:}1, x{:}1, y{:}2, y'{:}1, z{:}1, z'{:}2\})$

$= 9$

3a.ii

$\neg((A_L{=}11{<}9{=}B_L){\wedge}(A_S{=}5{\leq}4{=}B_S))$

$\neg((B_L{=}9{<}11{=}A_L){\wedge}(B_S{=}4{\leq}5{=}A_S))$

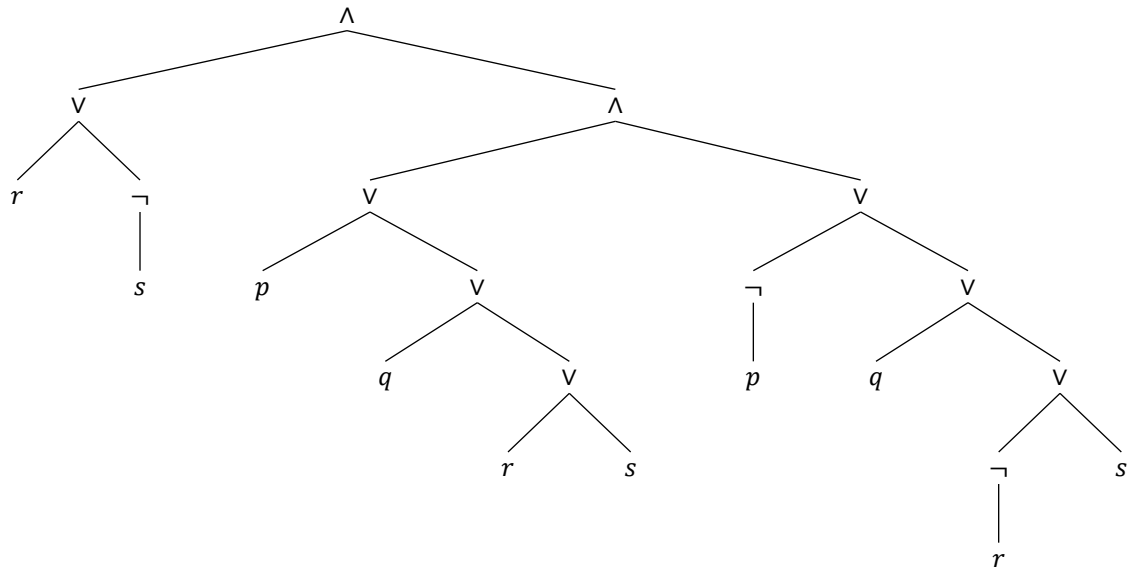
$Smplr(B,A)$

$$G(w,x,y,z) = (y + z')(w + x + y + z)(w' + x + y' + z)$$

$$\psi(p,q,r,s)$$

$$= (r \vee (\neg s)) \wedge (p \vee q \vee r \vee s) \wedge ((\neg p) \vee q \vee (\neg r) \vee s)$$

Tree



2 (3b) Find the versions: $H(A, B, C, D)$ of $G(w, x, y, z)$

and $K(A, B, C, D)$ of $MSOP\left(G(w, x, y, z)\right)$

in Sets.

1 (i) $H(A, B, C, D)$

$= (C \cup (D^c)) \cap (A \cup B \cup C \cup D) \cap ((A^c) \cup B \cup (C^c) \cup D)$

1 (i) $K(A, B, C, D)$

$= ((A^c) \cap C) \cup (C \cap D) \cup (B \cap (D^c)) \cup (A \cap (C^c) \cap (D^c))$

3b.i

$$G(w,x,y,z) = (y + z')(w + x + y + z)(w' + x + y' + z)$$

$$H(A,B,C,D) = (C \cup (D^{c})) \cap (A \cup B \cup C \cup D) \cap ((A^{c}) \cup B \cup (C^{c}) \cup D)$$

3b.ii

$$MSOP(G(w,x,y,z)) = w'y + wy'z' + xz' + yz$$

$$K(A,B,C,D) = ((A^{c}) \cap C) \cup (C \cap D) \cup (B \cap (D^{c})) \cup (A \cap (C^{c}) \cap (D^{c}))$$

9 (3c) Prove in Sets that:

$$H(A, B, C, D) = K(A, B, C, D)$$

The correspondence w,x,y,z to A,B,C,D together with complement to complement, $+$ to union, and juxtaposition to intersection is a Boolean-algebra isomorphism between BA and Sets (both are the free Boolean algebra on 4 generators). By definition, $H(A,B,C,D)$ is exactly this isomorphism applied to $G(w,x,y,z)$, and $K(A,B,C,D)$ is the same isomorphism applied to $MSOP(G(w,x,y,z))$.

In (2c), the consensus method established, in BA:

$$G(w,x,y,z) = \ w'y + wy'z' + xz' + yz$$

A Boolean-algebra isomorphism preserves every identity, so applying the same correspondence to both sides of this identity gives, in Sets:

$$H(A,B,C,D) = \ K(A,B,C,D)$$

as required.

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